Chapter 3 · Energy and Speed Are Related

Teacher Key · Guided Notes

Name: _____ **Date:** _____

BQ How are energy and speed related?

My first thinking

Any first idea is acceptable. Do not require the relationship yet.

V1 Vocabulary — Speed

Reader p. 13

speed — a measurement of the **distance** an object travels over an amount of time.

N1 One Ball, Two Speeds

Reader p. 13-14

Compare the **same** ball at two speeds.

A bigger **change** is how we know the faster ball had more energy of motion.

BALL → OBSERVE → EVIDENCE

Preview: one ball, two setups. Which cup move was bigger?

N2 What We Measured

Teacher Guide Lesson 4 · AP 4.1

We measure ramp **height** in centimeters.

We measure how far the **cup** moves in centimeters.

We do not measure a speed number.

The one thing we change is ramp **height**.

We keep the **same** ball.

Record the nine measured trials on the Ramp It Up sheet. The first roll is a calibration trial. Do not record it.

Which setup do you predict will move the cup farther? Why?

What change did you measure?

How far the cup moved, in centimeters. Not a speed number.

N3 Steeper Ramp

Reader p. 15

Objects pick up more **speed** on steeper ramps.

From the steeper ramp, that ball made a **greater** change.

N4 The Bounce

Reader p. 16

The faster ball bounced **higher** than the slower ball.

A louder sound is also a bigger change you can observe.

CHECK Your Thinking

How are energy and speed related? Name the ball you kept and one change that was bigger.

That ball had more energy of motion when it caused a greater change. I did not measure a speed number.

BQ · NOW I THINK How are energy and speed related?

For the same object, greater speed is related to greater energy of motion. A greater change is the evidence.